

One Ticker Deep: The Trend Strategy That Passed Every Test and Still Shouldn't Be Believed

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Companion paper to "Do Simple Cross-Sectional Trend Strategies Survive Honest Validation? Evidence from a Small US Large-Cap Universe, 2020–2024."

Abstract

In a companion study I showed that two canonical cross-sectional long-short strategies — moving-average crossover and 12-1 momentum — fail walk-forward validation on US large-caps over 2020–2024, and that a single train/test split overstates their performance roughly threefold. This paper studies the strategy that *passed*. A directional, long-only time-series trend rule — hold the strongest names trading above their own moving average, retreat to cash otherwise — survives the identical walk-forward protocol with an out-of-sample Sharpe of 1.36 on the original 26-name universe (CAGR 34.1% against 20.8% for the S&P 500), and 1.53 on a broadened 99-name universe selected on start-of-sample information. Every cell of its 36-point parameter grid beats buy-and-hold in-sample. By the standards of the companion paper, this strategy is validated. I then attribute the result, in three layers. Volatility terciles reveal that the performance is monotone in volatility and lives entirely in the highest-volatility third of the universe; in the lower two terciles the strategy underperforms buy-and-hold. Excluding a single stock — NVIDIA — from either universe collapses the walk-forward Sharpe below the S&P 500's (1.13 and 1.05 versus 1.30) and shrinks the CAGR edge from over thirteen percentage points to about two. Repairing the survivorship bias of the universe does not repair the concentration: the dependence on one historic outlier is a property of the strategy, not of the stock list. As a cross-check, the same broadened universe fails to rescue the long-short strategies (walk-forward Sharpes of -0.31 and 0.18), rejecting the breadth hypothesis advanced in the companion paper. What survives every test is trend-following's drawdown reduction — maximum drawdowns of -13% to -19% against -24% to -34% for buy-and-hold — at roughly market-like returns. The methodological conclusion extends the companion paper's: a clean validation protocol can still bless a lucky strategy. Validate, then attribute, before believing any backtest.

Keywords: trend following, time-series momentum, backtest overfitting, attribution, survivorship bias, concentration risk

JEL codes: G11, G12, G14

1. Introduction

The companion paper to this one ends with a defensible negative: neither a moving-average crossover nor classic cross-sectional momentum, implemented dollar-neutral on liquid US large-caps, survives walk-forward validation over 2020–2024. The natural question a practitioner asks next is whether the *validation pipeline* is simply too harsh — whether anything can pass it. This paper answers that question, and the answer turns out to be more instructive than either a plain yes or a plain no.

The candidate is the simplest directional strategy in the practitioner canon: time-series trend following (Moskowitz, Ooi and Pedersen, 2012; Faber, 2007). Each stock is compared to its own trailing moving average rather than to its peers. Names trading above their average are candidates to hold; names below it are sold. The portfolio holds the few strongest candidates and retreats to cash when trends die, so the book de-risks mechanically in broad downturns. Unlike the dollar-neutral constructions of the companion paper, this strategy deliberately carries market beta; the fair benchmark is therefore buy-and-hold itself, and the bar is to beat it.

By every standard applied in the companion paper, the strategy clears the bar. Its in-sample parameter surface is not a lucky spike but a uniform plateau: all 36 combinations of trend window and portfolio breadth beat the S&P 500's Sharpe ratio, with a median of 1.20 against 0.76. It survives the walk-forward protocol that dismantled the long-short strategies, posting an out-of-sample Sharpe of 1.36 with a fourteen-percentage-point CAGR advantage over the index. And when I repair the most serious data defect of the companion study — a universe hand-picked in 2026, after the sample — by re-selecting roughly one hundred names on January 2020 index membership, the walk-forward result *improves*, to 1.53.

A researcher who stopped here would deploy this strategy. The contribution of this paper is to show, in three attribution layers, why that would be a mistake — and what a more careful reading of the same evidence actually supports.

First, sorting the universe into volatility terciles shows the performance is monotone in volatility and concentrated entirely in the top tercile — the NVDA, META, AMZN cohort. In the bottom two terciles, covering two-thirds of the universe, the strategy *underperforms* buy-and-hold. The pattern replicates exactly on the survivorship-corrected universe. Whatever this strategy harvests, it is not a universal trend effect; it is a property of high-volatility growth names in this particular regime.

Second, removing a single stock — NVIDIA, which rose roughly tenfold over the sample — collapses the strategy's walk-forward Sharpe from 1.36 to 1.13 on the original universe and from 1.53 to 1.05 on the broadened one, in both cases below the S&P 500's 1.30 over the same span. The CAGR advantage shrinks from +13.3 and +20.5 percentage points to +2.2 and +2.4. One ticker out of twenty-six, or out of ninety-nine, accounts for essentially the entire risk-adjusted edge. Strikingly, the ex-NVDA CAGR is almost identical across the two universes (23.0% versus 23.2%): remove the outlier and the strategy converges to market-like returns regardless of the stock list.

Third, the broadened universe serves as a controlled experiment on the companion paper's leading conjecture — that the long-short strategies failed for lack of cross-sectional breadth. The conjecture is rejected: quadrupling the universe leaves momentum's walk-forward Sharpe essentially unchanged (0.18 versus 0.22) and pushes the MA long-short *negative* (−0.31 versus 0.35). Breadth was never the binding constraint; the signals were.

The residual finding, after all three layers, is modest and I believe genuine: the trend rule's cash-retreat mechanism reliably truncates drawdowns (−13% to −19% across every specification, against −24% to −34% for buy-and-hold) at roughly market-like returns. That is a defensible property with independent support in the literature (Faber, 2007; Moskowitz, Ooi and Pedersen, 2012). It is also a much smaller claim than the headline backtest appears to support.

The paper's method is its message. The companion study showed that the choice of validation protocol can fabricate an edge; this study shows that even a sound protocol can certify one that attribution immediately dissolves. Walk-forward validation asks *whether* a strategy would have worked. Attribution asks *why* — and a "why" of "one historic stock happened to be in the list" is not an edge, however clean the protocol that reported it.

Section 2 describes the data and the two universes. Section 3 specifies the strategy and recapitulates the shared evaluation machinery. Section 4 presents results in the order the experiments were run — baseline, parameter surface, validation, then the three attribution layers — and closes with a master comparison of every scenario generated. Section 5 discusses interpretation, Section 6 limitations, and Section 7 concludes. Appendix A contains the strategy and attribution source code; Appendix B gives replication instructions.

2. Data and universes

2.1 Source

All data are split- and dividend-adjusted daily bars from the Alpaca Market Data API, 2 January 2020 to 30 December 2024 (1,257 trading days), with fields open, high, low, close, volume, trade count, and daily VWAP. Data-integrity checks (adjustment verification at known splits; VWAP within [low, high] on 100% of rows) are described in the companion paper. SPY proxies buy-and-hold and is never tradable.

2.2 The two universes: a controlled comparison

The companion paper's universe — 26 liquid large-caps across five sectors — was hand-picked in 2026, i.e. *after* the sample. That is a textbook survivorship defect: the list contains the names known, with hindsight, to have remained large and liquid, and conspicuously includes NVIDIA. For a dollar-neutral book the defect is partially self-hedging (both the longs and the shorts come from the same flattered list). For a long-only strategy that concentrates in recent winners it is potentially fatal, which motivates the second universe.

The 2020 universe. Ninety-nine names reconstructing S&P 100 membership as of January 2020 — selection using nothing known after the sample begins. It therefore includes the subsequent losers a 2026 list quietly drops: Boeing, General Electric, Walgreens, Intel, Disney, PayPal, Citigroup, 3M and others. All 99 names have full 1,257-day coverage on Alpaca.

Two residual biases are documented rather than hidden. Members delisted mid-sample through corporate actions (Allergan, acquired May 2020; Raytheon/United Technologies, merged April 2020) could not be included because the data vendor does not serve delisted tickers; the list therefore still slightly overweights firms that survived 2020–2024 as independent companies (Shumway, 1997, documents the general form of this bias). And membership is reconstructed from public sources rather than a licensed point-in-time constituents file, so small errors are possible. Neither caveat affects the paper's central comparisons, which hold the universe fixed while varying the strategy, or vice versa.

3. Strategy and methodology

3.1 The trend rule

For stock i on day t , with an n -day simple moving average:

$$score_{it} = P_{it} / SMA_n(P_i)_t - 1$$

A positive score means the stock trades above its own n -day average — the time-series definition of an uptrend, as distinct from the cross-sectional ranking of the companion paper, in which the same arithmetic would be compared *across* stocks. Here the zero line carries meaning: it is the buy/sell boundary.

Each rebalance (every 5 trading days), among stocks with positive scores, the `n_hold` strongest are held at fixed weight $1/n_hold$ each. If fewer than `n_hold` names qualify, the shortfall remains in cash; if none qualify, the book is entirely in cash. Sizing is deliberately fixed rather than renormalised, so gross exposure floats between 0 and 100% with the breadth of the uptrend — this is the mechanism that de-risks the book in broad selloffs without any dedicated crash logic. Positions are held constant between rebalances. The default configuration is a 100-day window holding 5 names; both parameters are swept in Section 4.2.

3.2 Shared machinery

The backtest engine, cost model and validation protocols are identical to the companion paper, to which I refer for detail: no look-ahead (weights lag their signal by one day), costs of 5 bps per unit of gross turnover, close-to-close accounting, Sharpe with the risk-free rate at zero (noted as a convention; for a long-only book this flatters slightly more than for a dollar-neutral one). The two out-of-sample protocols are likewise unchanged: a 60/40 chronological train/test split, and a walk-

forward that re-selects parameters every 126 trading days on a trailing 378-day window, concatenating the out-of-sample blocks. The parameter grid for selection is $\text{window} \in \{20, 40, 60, 80, 100, 125, 150, 200, 250\} \times \text{n_hold} \in \{3, 5, 8, 10\}$, 36 combinations.

3.3 Attribution protocols

Three attribution layers, fixed before inspecting their results:

Volatility terciles. Rank the universe by full-sample annualised volatility, split into thirds, and run the strategy independently within each tercile. Using full-sample volatility involves mild look-ahead (rankings are, however, highly persistent); the diagnostic purpose is to locate the performance, not to propose a tradable sort.

Single-name exclusion. Re-run the entire walk-forward — selection and all — on the universe minus its largest single contributor, NVIDIA. If a strategy's edge is a portfolio property, removing one of 26 (or one of 99) names should dent it modestly; if the edge collapses, the "portfolio strategy" was a single-stock bet wearing a systematic costume.

Universe replacement. Re-run the walk-forward on the 2020 universe, with and without NVIDIA, and — as a cross-check on the companion paper's breadth conjecture — re-run the two long-short strategies on the broadened universe as well.

4. Results

4.1 The baseline looks excellent

Table 1 reports the default configuration (100-day window, 5 names) on the 26-name universe, against buy-and-hold.

Table 1. Trend-following baseline versus SPY, 26-name universe, 2020–2024

Metric	Trend-follow 100d/5	SPY buy & hold
Total return	184.1%	95.3%
CAGR	23.3%	14.4%
Sharpe	1.12	0.76
Maximum drawdown	–23.2%	–33.8%
Annualised volatility	20.6%	20.6%
Average gross exposure	91.2%	100%

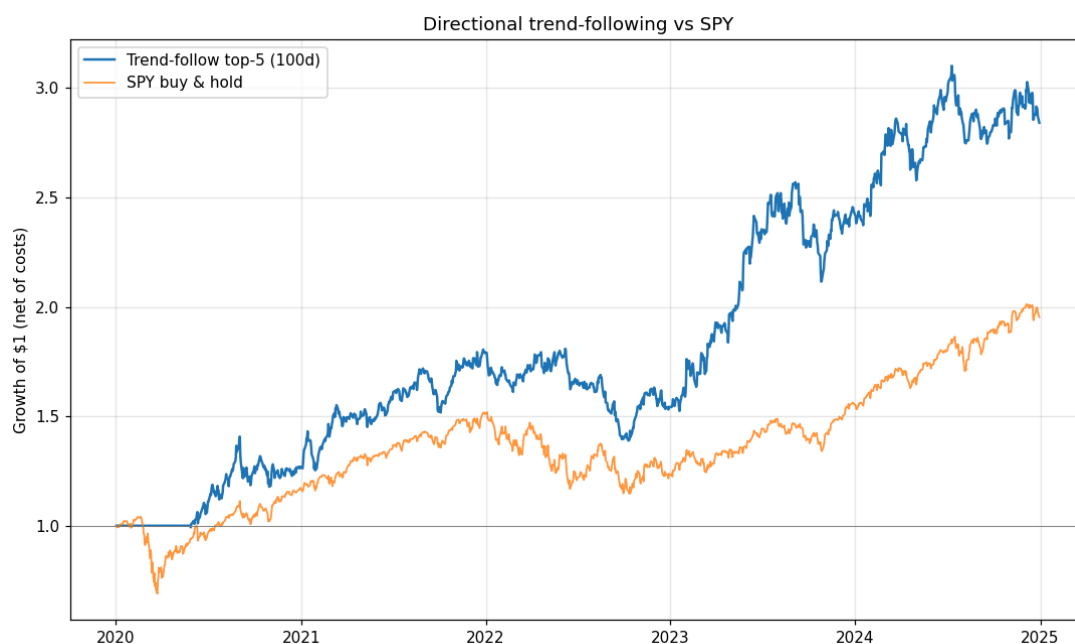


Figure 1. Baseline trend-following versus SPY, 2020–2024. The strategy is flat through the February–March 2020 crash — the cash-retreat mechanism operating as designed — then rides the recovery.

At identical volatility the strategy earns nearly twice the index's CAGR with a third less drawdown. The equity curve shows why: the book was in cash through the COVID crash (partly warm-up, partly no names above trend), then fully invested through the recovery. Where the companion paper's baseline opened with a failure, this one opens with apparent success — which is precisely why the rest of the paper exists.

4.2 The parameter surface is uniformly green — which should raise suspicion, not confidence

Sweeping the full grid (window $\times$ n_hold, 36 cells) yields the most striking in-sample table of either paper: **every cell beats buy-and-hold**. Sharpe ranges from 0.89 to 1.43 with a median of 1.20, against SPY's 0.76.

Table 2. In-sample sweep extremes, 26-name universe (36 cells; SPY Sharpe 0.76)

Cell	Sharpe	CAGR	Max DD	Ann. vol	Turnover
Best: 80d / 3 names	1.43	39.2%	–19.9%	25.3%	0.091
Worst: 20d / 3 names	0.89	21.3%	–35.7%	25.3%	0.215
Median of grid	1.20	—	—	—	—

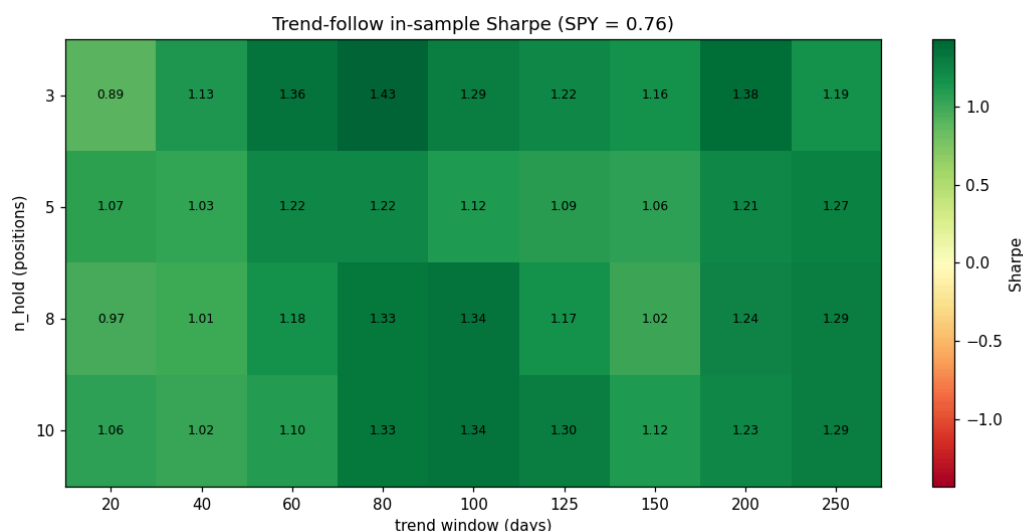


Figure 2. In-sample Sharpe by (window, n_hold). All 36 cells exceed SPY's 0.76. Concentration (3 names) maximises return at deeper drawdowns; breadth (8–10 names) smooths. The 20-day column is uniformly the worst.

Two structural observations. First, the horizon lesson of the companion paper recurs for the fifth time across these studies: the 20-day column is the worst on the surface, and the comparison of the two extreme cells isolates the mechanism. Both run identical volatility (25.3%), but the 20-day cell earns barely half the return with 2.4 times the turnover — it whipsaws, selling after dips and re-buying after rebounds, so it suffers a *worse* maximum drawdown than buy-and-hold (–35.7% versus –33.8%) despite being a "risk-managed" strategy. Days-scale signals are noise; the viable trend horizon is months.

Second, and more importantly: a 36-for-36 sweep on a universe selected after the sample should be read as a red flag, not a robustness certificate. When every variant of a long-only "hold the strongest" rule beats the market on a list containing the era's biggest winners, the natural hypothesis is that the *list*, not the rule, is doing the work. Sections 4.4–4.6 test that hypothesis directly.

4.3 The strategy survives both validation protocols

Train/test split. Selection on the training window (December 2020 – May 2023) picks 80/3 (train Sharpe 1.52). Frozen on the test window it earns a Sharpe of 1.20, CAGR 31.9%, maximum drawdown –19.9% (test-optimal oracle: 250/10 at 2.16).

Walk-forward. Table 3 reports the five folds.

Table 3. Walk-forward folds, 26-name universe

Test window	Selected (window/n_hold)	Fold Sharpe
Jun 2022 – Dec 2022	125/3	−0.69
Dec 2022 – Jun 2023	200/3	3.03
Jul 2023 – Dec 2023	80/3	0.83
Jan 2024 – Jul 2024	40/3	3.26
Jul 2024 – Dec 2024	40/3	−0.61
Stitched (629 days)		1.36

The stitched out-of-sample record: **Sharpe 1.36, CAGR 34.1%, total +108.0%, maximum drawdown −18.6%**, against SPY's 1.30 and 20.8% over the identical span.

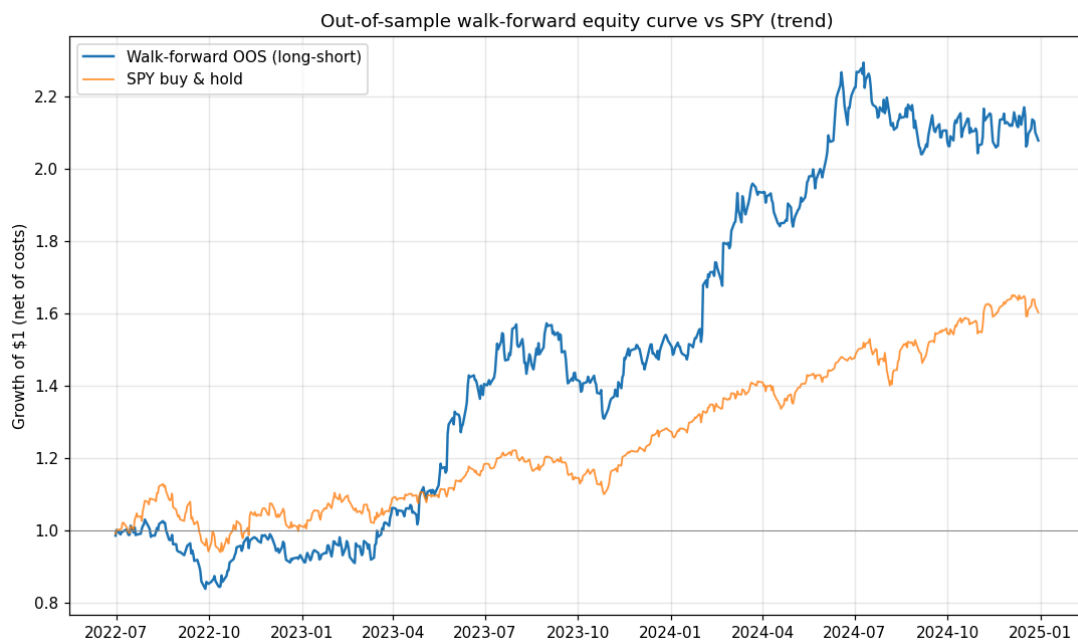


Figure 3. Stitched walk-forward out-of-sample equity curve versus SPY, 26-name universe. The first strategy in either paper to beat buy-and-hold out-of-sample.

This is the exact protocol that reduced the companion paper's strategies to Sharpes of 0.35 and 0.22. Here it certifies a fourteen-point CAGR advantage with a smaller drawdown. Three cautionary details temper the headline even before attribution. The Sharpe advantage itself is thin — 1.36 versus 1.30 over two and a half years is statistically indistinguishable; the genuine edge is the return-drawdown combination. Fold dispersion remains enormous (−0.69 to +3.26), with two of five folds negative. And while the selected *window* lurches between folds (125 → 200 → 80 → 40), the selected *breadth* never does: every fold chooses the most concentrated book allowed, `n_hold = 3`. The walk-forward is persistently maximising concentration — a detail that foreshadows everything that follows.

4.4 Attribution I: the performance is monotone in volatility

Table 4 runs the strategy independently within volatility terciles of each universe (trend rule at 100d/top-3; the companion paper's MA long-short within the same buckets for contrast).

Table 4. Performance within volatility terciles, full sample

Bucket	Trend Sharpe	Trend CAGR	MA L/S Sharpe	Trend Sharpe	Trend CAGR	MA L/S Sharpe
	<i>26-name universe</i>			<i>99-name universe</i>		
Low vol	0.60	7.0%	0.08	0.52	6.9%	-0.34
Medium vol	0.84	14.3%	-0.65	0.71	12.4%	-0.41
High vol	1.52	43.2%	0.76	1.22	36.6%	0.16
SPY	0.76	14.4%	—	0.76	14.4%	—

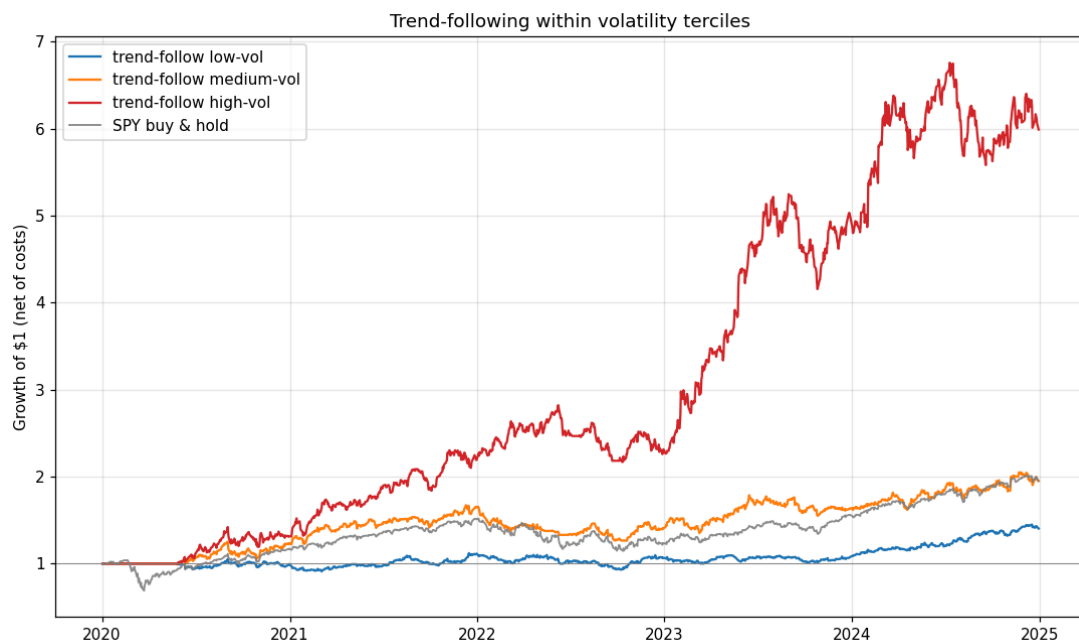


Figure 4. Trend-following within volatility terciles, 26-name universe. Low-vol tercile: JNJ, KO, PEP, WMT, MRK, MCD, PFE, V (annualised vol 19.7–27.9%). High-vol tercile: XOM, CVX, AMZN, BAC, ADBE, CRM, INTC, META, NVDA (34.2–53.9%).

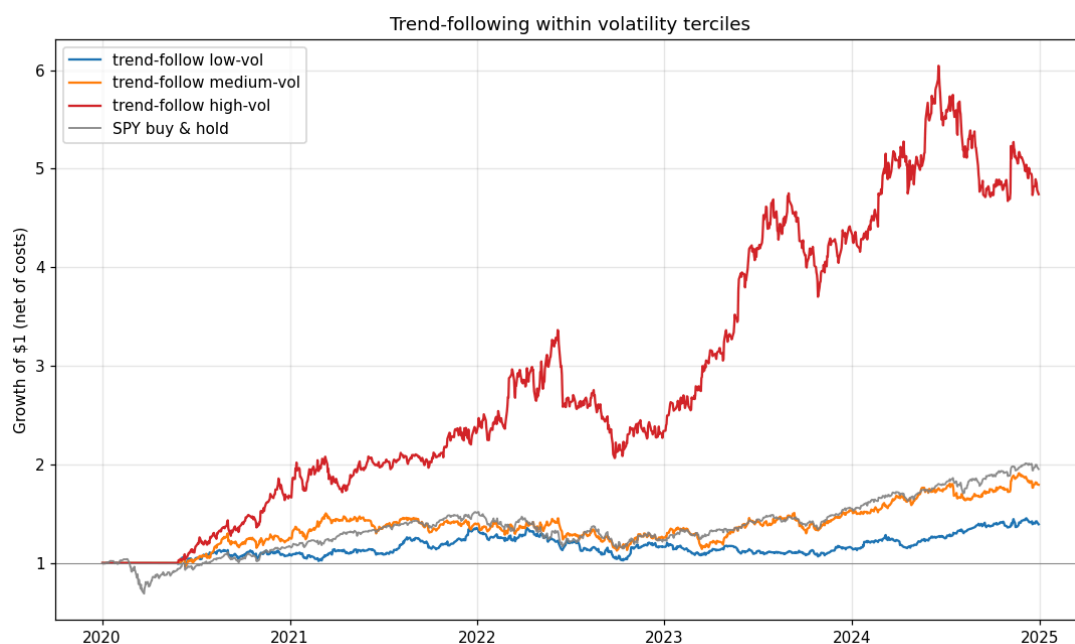


Figure 5. The same experiment on the 99-name 2020 universe (33 names per tercile). The monotone pattern replicates; the high-vol bucket's deeper drawdown ($\sim 38.6\%$) reveals downside the 2026-picked list had erased.

The pattern is identical on both universes and admits no ambiguity: performance rises monotonically with volatility, and the effect lives entirely in the top tercile. In the low and medium terciles — two-thirds of either universe — trend-following *underperforms* buy-and-hold. "Trend-following worked in 2020–2024" is the wrong description of these tables. "Trend-following high-volatility growth stocks worked in 2020–2024" is the right one, and it is a materially weaker claim: it identifies the return engine as a specific, regime-dependent cohort, not a universal effect. The broad-universe replication adds one sobering detail: the clean universe's high-vol tercile suffered a -38.6% maximum drawdown — worse than buy-and-hold — where the 2026-picked list showed only -23.2% . The survivorship-tainted list had erased not just losers but the true downside of the winners' cohort.

The volatility sort also suggests *why* every fold of the walk-forward selects maximal concentration: the fewer names the book holds, the higher the share of the era's few explosive winners. Which raises the obvious question of how much of the edge is attributable to the single most explosive one.

4.5 Attribution II: remove one stock and the edge is gone

Table 5 re-runs the complete walk-forward — the selection loop included, so the protocol has every chance to adapt — on each universe with NVIDIA excluded.

Table 5. Walk-forward out-of-sample, with and without NVDA

Universe	With NVDA		Without NVDA		SPY
	Sharpe	CAGR	Sharpe	CAGR	Sharpe / CAGR
26-name	1.36	34.1%	1.13	23.0%	1.30 / 20.8%
99-name (2020)	1.53	41.3%	1.05	23.2%	1.30 / 20.8%

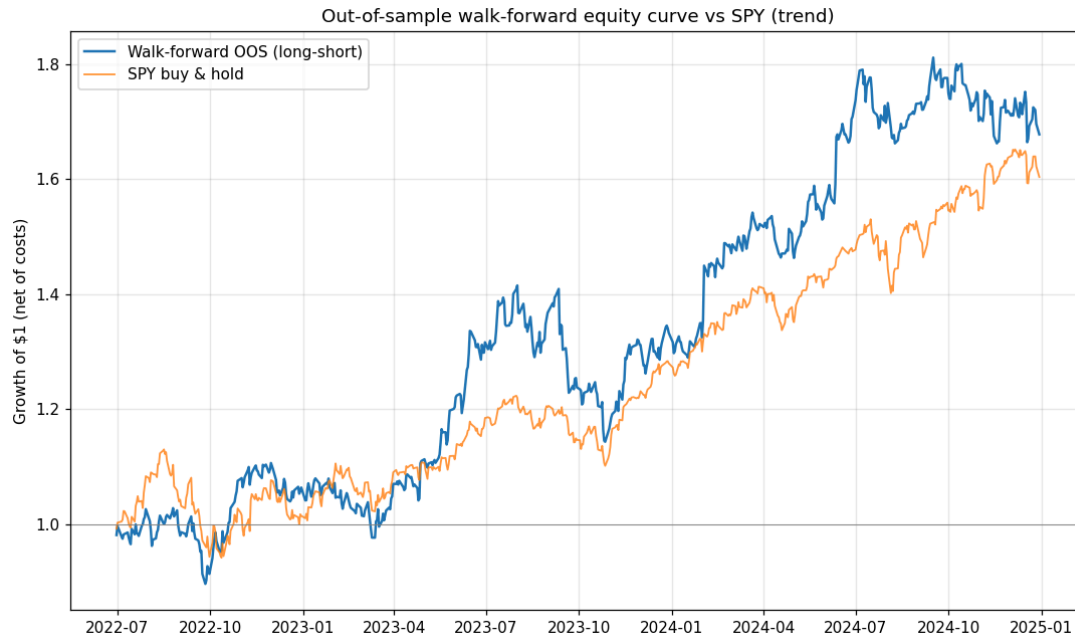


Figure 6. Walk-forward out-of-sample equity curve, 26-name universe excluding NVDA.

Removing one ticker — out of twenty-six, or out of ninety-nine — moves the strategy from "beats the index on every metric" to "risk-adjusted performance below the index" (1.13 and 1.05 versus 1.30). The CAGR advantage collapses from +13.3 and +20.5 percentage points to +2.2 and +2.4 — inside the noise band of a two-and-a-half-year window. Two further details deserve emphasis.

First, the ex-NVDA CAGRs are nearly identical across universes: 23.0% on 26 names, 23.2% on 99. Remove the outlier and the strategy converges to essentially the same market-like return regardless of the stock list — about as clean a demonstration as the data permit that the excess return *was* NVIDIA.

Second, the drawdown improvement survives the exclusion: −19.2% and −16.7% ex-NVDA, against roughly −24% for the index over the same span. What the exclusion destroys is the alpha claim; what it leaves intact is the risk-management claim.

4.6 Attribution III: the universe fix does not fix the strategy — and rejects the breadth conjecture

The 99-name results above double as the survivorship experiment. Note the direction of the effect: on the *cleaner* universe, the with-NVDA result got *better* (1.53 versus 1.36; CAGR 41.3%, drawdown −13.4%, no negative folds), because 73 additional 2020-era names gave the selector more high-vol

winners to rotate through. The survivorship correction, by itself, would have *increased* confidence in the strategy. Only the exclusion test reveals that the improved number rests on the same single-name foundation ($1.53 \rightarrow 1.05$). Concentration risk is a property of the strategy — a top-3 book will find NVIDIA in any universe that contains it — and no amount of universe hygiene removes it.

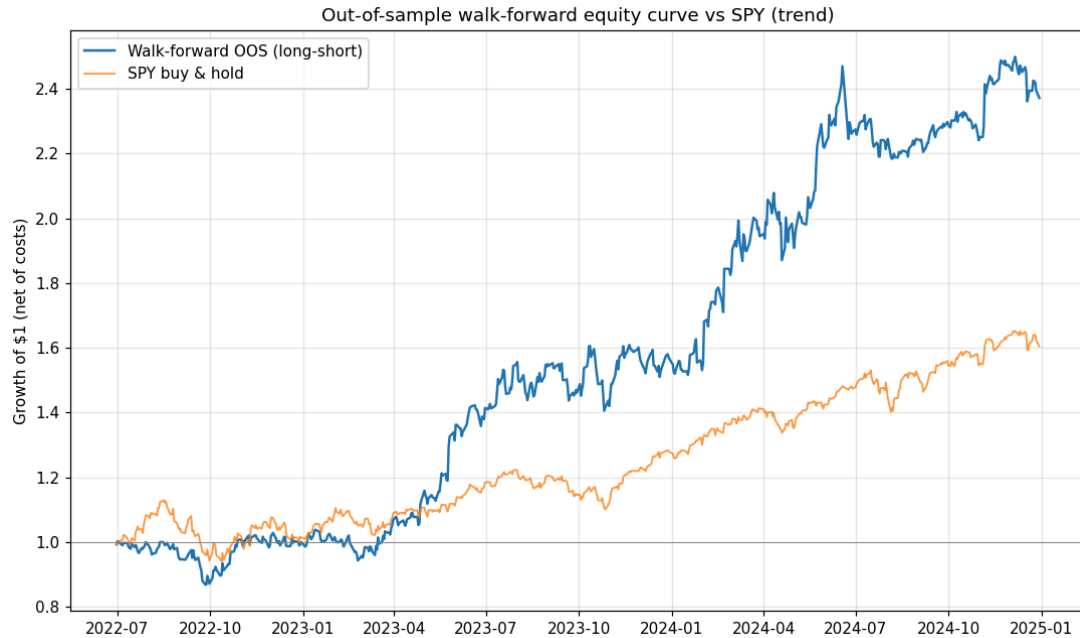


Figure 7. Walk-forward out-of-sample equity curve, 99-name 2020 universe (with NVDA): the best out-of-sample result of either paper — until attribution.

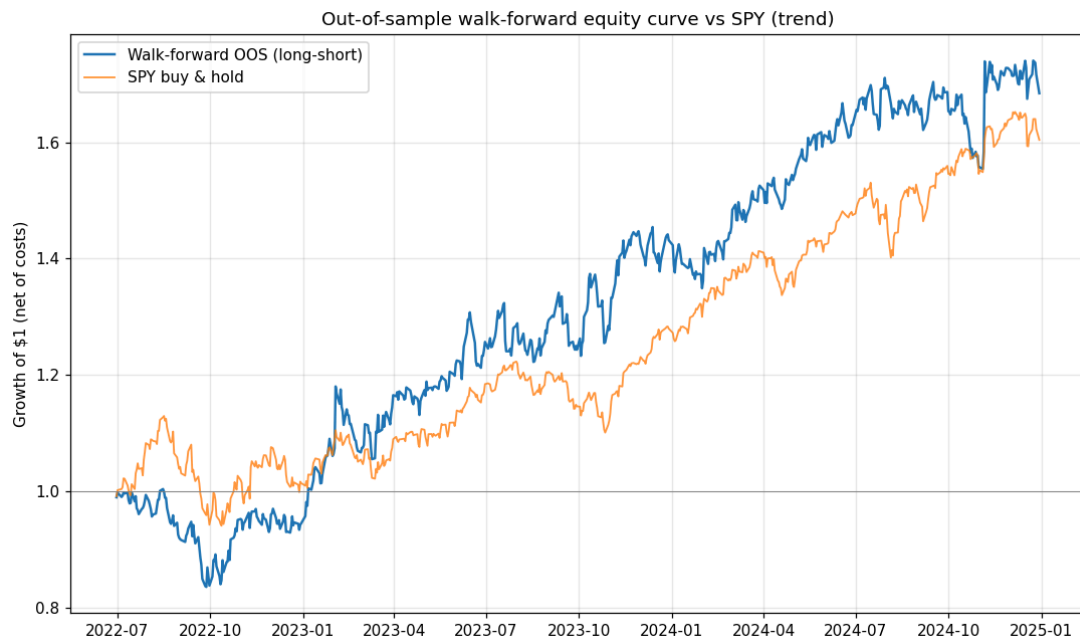


Figure 8. The same experiment excluding NVDA. The out-of-sample edge over SPY essentially disappears.

Finally, the broadened universe adjudicates the companion paper's leading conjecture — that its long-short strategies failed for want of cross-sectional breadth, momentum's documented edge having been established on universes of hundreds to thousands of names. Table 6 reports the same walk-forward protocol for all three strategies on both universes.

Table 6. Walk-forward out-of-sample Sharpe, all strategies × both universes

Strategy	26-name (2026-picked)	99-name (2020-selected)	SPY
MA crossover long-short	0.35	-0.31	1.30
Momentum (12-1) long-short	0.22	0.18	1.30
Trend-following (directional)	1.36	1.53	1.30
Trend-following ex-NVDA	1.13	1.05	1.30

The conjecture is rejected. Quadrupling the universe — with the 2022 losers included, and hence *more* genuine cross-sectional dispersion — leaves momentum essentially unchanged (0.18 versus 0.22) and pushes the MA long-short from marginally positive to outright negative (-0.31 versus 0.35), revealing its small-universe result as fragile in its own right. Neither long-short signal has an out-of-sample edge at either universe size. The binding constraint was never breadth; it was the signals themselves, in this regime. (One construction caveat: the long-short book was held at 5 names per side on both universes for comparability — 19% tails on 26 names but 5% tails on 99. A decile-style construction on the broad universe was not tested and could in principle behave differently.)

4.7 The complete scenario comparison

Table 7 collects every scenario generated across the two papers into a single view. In-sample rows are shaded by the caveat that applies to them; all walk-forward rows share the identical protocol and OOS span (mid-2022 to end-2024, 627–629 trading days).

Table 7. Master comparison of all scenarios

#	Scenario	Universe	Type	Sharpe	CAGR	Max DD
1	Trend 100d/5 baseline	26	in-sample	1.12	23.3%	−23.2%
2	Trend sweep, best cell (80/3)	26	in-sample (max of 36)	1.43	39.2%	−19.9%
3	Trend sweep, worst cell (20/3)	26	in-sample (min of 36)	0.89	21.3%	−35.7%
4	Trend, train/test split	26	OOS (single split)	1.20	31.9%	−19.9%
5	Trend, train/test split	99	OOS (single split)	1.36	41.3%	−18.1%
6	Trend, walk-forward	26	OOS (walk-forward)	1.36	34.1%	−18.6%
7	Trend, walk-forward	99	OOS (walk-forward)	1.53	41.3%	−13.4%
8	Trend, walk-forward, ex-NVDA	26	OOS (walk-forward)	1.13	23.0%	−19.2%
9	Trend, walk-forward, ex-NVDA	99	OOS (walk-forward)	1.05	23.2%	−16.7%
10	Trend, low-vol tercile	26 / 99	in-sample	0.60 / 0.52	7.0% / 6.9%	−16.9% / −24.4%
11	Trend, medium-vol tercile	26 / 99	in-sample	0.84 / 0.71	14.3% / 12.4%	−24.4% / −25.1%
12	Trend, high-vol tercile	26 / 99	in-sample	1.52 / 1.22	43.2% / 36.6%	−23.2% / −38.6%
13	MA long-short, walk-forward	26	OOS (walk-forward)	0.35	3.8%	−15.2%
14	MA long-short, walk-forward	99	OOS (walk-forward)	−0.31	−5.4%	−33.5%
15	Momentum 12-1, walk-forward	26	OOS (walk-forward)	0.22	2.1%	−23.3%
16	Momentum 12-1, walk-forward	99	OOS (walk-forward)	0.18	1.7%	−20.9%
17	SPY buy & hold, OOS span	—	benchmark	1.30	20.8%	~−24%

#	Scenario	Universe	Type	Sharpe	CAGR	Max DD
18	SPY buy & hold, full sample	—	benchmark	0.76	14.4%	−33.8%

Read vertically, the table is the paper's argument. Rows 1–7 build an apparently overwhelming case: a robust plateau, two passed validation protocols on two universes, culminating in row 7's Sharpe of 1.53 with a −13.4% drawdown. Rows 8–12 dismantle it: the same protocol minus one stock (rows 8–9) drops below the benchmark, and the tercile decomposition (rows 10–12) locates the entire effect in one-third of the universe. Rows 13–16 show the long-short alternatives were worse everywhere. What remains constant down the entire trend-following block — with or without NVDA, on either universe — is the drawdown column sitting well above the benchmark's.

5. Discussion

5.1 Why validation certified a lucky strategy

The walk-forward protocol did nothing wrong. At every fold it used only trailing information; its selections were reasonable; its out-of-sample accounting was clean. It certified the strategy because, within the data it was shown, the strategy genuinely would have worked. The failure is one of *scope*: the protocol answers "would this rule, applied to this universe, have beaten the index?" — a question whose answer can be dominated by a single constituent. NVIDIA's ascent was real, and a concentrated trend rule genuinely would have ridden it. What the protocol cannot say is whether "a stock like NVIDIA exists in the universe and the strategy concentrates into it" is a repeatable mechanism or a one-draw event. That is an attribution question, and it required exactly three cheap experiments to answer: sort by the suspected driver (volatility), delete the suspected driver (NVDA), and change the sampling frame (the 2020 universe). Each experiment reuses the existing pipeline; together they took minutes of compute. There is no methodological excuse for omitting them, and the literature's warnings about multiple testing (Harvey, Liu and Zhu, 2016; Bailey and López de Prado, 2014) apply with equal force to this subtler, single-asset form of luck.

5.2 What the strategy actually is

The three layers agree on a description. The rule is a momentum-concentration device: it channels the portfolio into whatever the universe's highest-volatility uptrend is, and 2020–2024 supplied a historic one. Its Sharpe advantage is an artefact of that supply. Its drawdown advantage, by contrast, is mechanical — the cash-retreat triggers whenever broad trends break, which is a property of the rule, not of any particular stock — and accordingly it survives every attribution layer: −18.6%, −13.4%, −19.2%, −16.7% across specifications, against −24% to −34% for buy-and-hold. This aligns with the established reading of time-series momentum as primarily a tail-risk modifier (Moskowitz, Ooi and Pedersen, 2012; Faber, 2007): the honest residual claim is "market-like returns with materially smaller drawdowns," not "34% a year."

It is worth stating what would have to be true for the bigger claim to stand. The strategy would need high-volatility trend winners to exist, be identifiable *ex ante*, and recur across regimes — in effect, a stationary supply of NVIDIAs. The sample contains one such episode, which is exactly enough to produce these backtests and exactly too few to support the claim.

5.3 The breadth conjecture, and the value of negative controls

The companion paper closed by conjecturing that its long-short failures reflected universe breadth. Table 6 rejects this, and the rejection has independent value: it was produced by a *negative control* — re-running unchanged strategies on the changed universe. Had the broadened universe rescued momentum, the survivorship fix would have been confounded with the breadth effect; because momentum stayed dead while trend-following improved, the two effects separate cleanly. Designing universe changes so that they test more than one hypothesis at a time is cheap, and this paper benefited from it twice.

6. Limitations

The attribution toolkit here is deliberately elementary — sorts, exclusions, and reruns — and stops short of formal factor attribution; regressing the strategy's returns on market, size, momentum and volatility factors would quantify what the sorts only locate. The volatility terciles use full-sample volatility (mild look-ahead, diagnostic only). The 2020 universe cannot include mid-sample delistings, leaving residual survivorship of the Shumway (1997) type. Costs remain 5 bps on turnover with no borrow, impact, or capacity modelling — generous to all strategies, though least distorting for the low-turnover trend rule (average daily turnover 0.09). The walk-forward has five folds; fold-level dispersion (−0.69 to +3.26) implies wide error bands on every stitched number, including the benchmark comparisons. All results concern one five-year window containing exactly one regime cycle. And the exclusion test removes only the single largest contributor; a top- k exclusion curve ($k = 1, 2, 3, \dots$) would trace the concentration profile more completely and is left for the replication package.

None of these limitations rescues the headline claim; each either weakens it further or leaves it unchanged. The drawdown-reduction finding is the one result robust to every variation tested.

7. Conclusion

A directional trend-following rule passed, with margin, the validation gauntlet that eliminated two canonical long-short strategies in the companion paper: every parameter cell beat buy-and-hold in-sample, and walk-forward out-of-sample Sharpes reached 1.36 on the original universe and 1.53 on a survivorship-corrected one. Three attribution layers then reduced the claim to its true size: the performance is confined to the highest-volatility tercile, collapses below the benchmark when one stock is removed from either universe, and improves with universe hygiene only because a cleaner

list supplies the same concentration mechanism with more fuel. The durable residual is a modest, well-precedented property — market-like returns with maximum drawdowns roughly a third smaller than buy-and-hold — plus a rejected conjecture: universe breadth does not resurrect the long-short signals.

The pair of papers now states a two-part discipline. The first showed that a favourable *protocol* can fabricate out-of-sample evidence: one split reported triple the walk-forward's Sharpe. This one shows that a sound protocol can still certify a lucky *composition*: the walk-forward's approval was real, and one ticker deep. Validation asks whether a strategy would have worked; attribution asks why; and a backtest deserves belief only after both questions have answers that survive each other.

References

- Bailey, D. H., Borwein, J. M., López de Prado, M., and Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. *Notices of the American Mathematical Society*, 61(5), 458–471.
- Bailey, D. H., and López de Prado, M. (2014). The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- Barroso, P., and Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Brock, W., Lakonishok, J., and LeBaron, B. (1992). Simple technical trading rules and the stochastic properties of stock returns. *Journal of Finance*, 47(5), 1731–1764.
- Daniel, K., and Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Faber, M. T. (2007). A quantitative approach to tactical asset allocation. *Journal of Wealth Management*, 9(4), 69–79.
- Harvey, C. R., Liu, Y., and Zhu, H. (2016). ...and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Jegadeesh, N., and Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- Malan, P. (2026). Do simple cross-sectional trend strategies survive honest validation? Evidence from a small US large-cap universe, 2020–2024. Working paper, SSRN.
- Moskowitz, T. J., Ooi, Y. H., and Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2), 228–250.
- Shumway, T. (1997). The delisting bias in CRSP data. *Journal of Finance*, 52(1), 327–340.

Sullivan, R., Timmermann, A., and White, H. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance*, 54(5), 1647–1691.

Appendix A. Source code

The engine, cost model and walk-forward loop are listed in the companion paper's appendix and are unchanged; the components below are what this paper adds. Repository layout: `research/{data, strategies, backtest}`.

A.1 The trend-following strategy (`research/strategies/trend_follow.py`)

```
"""Directional (long-only) trend following – time-series momentum.

Each stock is judged against its OWN trend: score = price / SMA(window) - 1.
At each rebalance we hold the `n_hold` strongest names among those in an
uptrend (score > 0), 1/n_hold weight each. If fewer than n_hold names are
trending up, the remainder stays in cash – so the book de-risks automatically
in broad downturns."""

import numpy as np
import pandas as pd

def trend_score(prices: pd.DataFrame, window: int = 100) -> pd.DataFrame:
    return prices / prices.rolling(window).mean() - 1.0

def weights(prices, window=100, n_hold=5, rebalance_days=5):
    score = trend_score(prices, window)

    target = pd.DataFrame(np.nan, index=prices.index, columns=prices.columns)
    rebal_dates = prices.index[window::rebalance_days]
    for dt in rebal_dates:
        row = score.loc[dt].dropna()
        up = row[row > 0].sort_values(ascending=False)
        w = pd.Series(0.0, index=prices.columns)
        if len(up) > 0:
            picks = up.index[:n_hold]
            w[picks] = 1.0 / n_hold      # fixed sizing; shortfall stays in cash
        target.loc[dt] = w
    return target.ffmpeg().fillna(0.0)    # hold between rebalances
```

A.2 Volatility terciles (research/run_vol_buckets.py , core)

```
def bucket_universe(prices):
    """Split the universe into volatility terciles (full-sample annualised
    vol; mild look-ahead, acceptable for a diagnostic sort)."""
    vol = (prices.pct_change().std() * np.sqrt(252)).sort_values()
    n = len(vol)
    return {
        "low":    list(vol.index[: n // 3]),
        "medium": list(vol.index[n // 3: 2 * n // 3]),
        "high":   list(vol.index[2 * n // 3:]),
    }

# Within each bucket the strategies are run unchanged on the sub-panel:
# prices_bucket = all_prices[buckets[name]]
# engine.run(prices_bucket, trend_follow.weights(prices_bucket, 100, 3, 5))
```

A.3 Exclusion and universe switching (research/validation.py , argument handling)

```
# Optional args after the strategy name:
#   broad      -> use the 99-name start-of-2020 universe (survivorship fix)
#   NVDA,META  -> exclude tickers (single-name attribution)
excl, use_broad = set(), False
for a in sys.argv[2:]:
    if a.lower() == "broad":
        use_broad = True
    else:
        excl |= set(a.upper().split(","))
base = UNIVERSE_2020 if use_broad else UNIVERSE
universe = [s for s in base if s not in excl]
```

The walk-forward itself (grid selection on a trailing 378-day window, traded over the following 126 days, out-of-sample blocks concatenated) is identical to the companion paper's Appendix A.4, with the trend strategy's grid being $\text{window} \in \{20, 40, 60, 80, 100, 125, 150, 200, 250\} \times \text{n_hold} \in \{3, 5, 8, 10\}$.

A.4 The 2020 universe (`research/universe_2020.py`, abridged)

```
"""Broad universe selected on START-OF-SAMPLE information (survivorship fix).

Approximate S&P 100 membership as of January 2020, reconstructed. Selection
uses nothing known after 2020-01, so subsequent losers (BA, GE, WBA, INTC,
DIS, PYPL, ...) are IN. Documented residual biases: members delisted
mid-sample via M&A (AGN; RTN/UTX -> RTX) unavailable from the data vendor;
membership reconstructed from public sources, not a licensed point-in-time
file."""

UNIVERSE_2020 = [
    "AAPL", "ABBV", "ABT", "ACN", "ADBE", "AIG", "ALL", "AMGN", "AMT", "AMZN",
    # ... 79 further names ...
    "UNP", "UPS", "USB", "V", "VZ", "WBA", "WFC", "WMT", "XOM",
] # 99 tickers, all with full 1,257-day coverage
```

Appendix B. Replication

With Alpaca paper-trading keys in a local `.env` (see companion paper):

```
python research/data/fetch_bars.py # 26-name universe + SPY
python research/data/fetch_bars.py $(python -c "from research.universe_2020 \
import UNIVERSE_2020; print(' '.join(UNIVERSE_2020))") # 99-name universe

python research/run_trend_follow.py # Table 1, Figure 1
python research/sensitivity_trend.py # Table 2, Figure 2
python research/validation.py trend # Table 3, Figure 3
python research/run_vol_buckets.py # Table 4 left, Figure 4
python research/run_vol_buckets.py broad # Table 4 right, Figure 5
python research/validation.py trend NVDA # Table 5 row 1, Figure 6
python research/validation.py trend broad # Table 5 row 2, Figure 7
python research/validation.py trend broad NVDA # Table 5 row 2, Figure 8
python research/validation.py ma broad # Table 6
python research/validation.py mom broad # Table 6
```

Figures are written to `research/backtest/:` `trend_follow.png`, `trend_follow_sweep.png`,
`walk_forward_oos_trend.png`, `vol_buckets.png`, `vol_buckets_broad.png`,
`walk_forward_oos_trend_ex_NVDA.png`, `walk_forward_oos_trend_broad.png`,
`walk_forward_oos_trend_broad_ex_NVDA.png`.